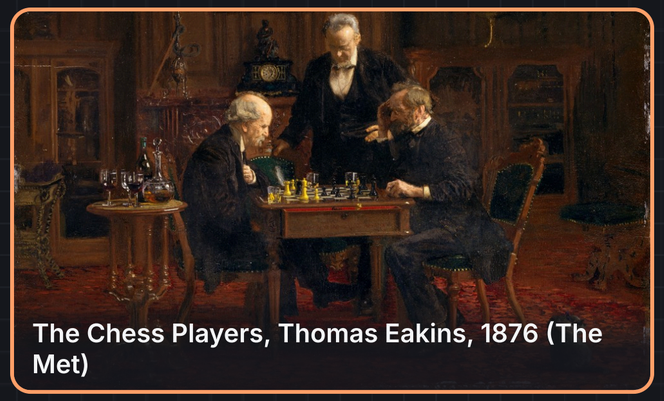
Pi & Planck

GAME THEORY

The Prisoner's Dilemma

Why a tournament of computer programs rewarded kindness.

WHERE IT CAME FROM



The Chess Players, Thomas Eakins, 1876 (The Met)

1950 Merrill Flood and Melvin Dresher devise the game at RAND.

1950 Albert Tucker tells it with prison sentences.

1950 John Nash proves equilibrium points exist (PNAS).

THE GAME

YOU	OTHER PLAYER: Cooperate	OTHER PLAYER: Defect
Cooperate	3 , 3	0 , 5
Defect	5 , 0	1 , 1

Whatever the other does, defecting pays you more. Yet both defecting (1, 1) is worse than both cooperating (3, 3).

200 MOVES, TOTAL POINTS



AXELROD'S TOURNAMENTS

Round 1
14 entries plus a random program. TIT FOR TAT won with 504.5 points.

Round 2
62 entries from six countries. TIT FOR TAT won again.

THREE TRAITS OF THE WINNER

Nice	never the first to defect
Provocable	answers a defection at once
Forgiving	cooperates again after one reply

Round-one scores: all of the top eight were nice.



5 · 3 · 1 · 0 points — Temptation, mutual cooperation, mutual defection, and the sucker's payoff.

504.5 points — TIT FOR TAT's round-one score. Cooperating for all 200 moves earns 600.

62 entries — Programs in round two. The simple rule won again.

Sources: R. Axelrod, J. Conflict Resolution 24 (1980) 3-25 and 379-403 · Axelrod and Hamilton, Science 211 (1981) 1390-1396 · J. Nash, PNAS 36 (1950) 48-49 · S. Kuhn, Stanford Encyclopedia of Philosophy · Image: The Metropolitan Museum of Art (public domain)

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